

Estimating Temperature of Grasped Object using PPG Sensor

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ABSTRACT

We propose a method to estimate the temperature of an object being grasped using a PPG sensor. Evaluation experiments showed a 94% accuracy for three subjects in two temperature ranges.

CCS CONCEPTS

• Human-centered computing → Ubiquitous computing.

KEYWORDS

Object temperature estimation, PPG sensor, pulse wave, wearable

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1 INTRODUCTION

Wearable devices such as smartwatches are widely used. Wearable devices can measure biometric information to obtain various types of information, such as pulse waves, and are applied to various applications, such as healthcare. In this paper, we propose a method that estimates the temperature of the grasped object from a pulse wave. It can be used for people with congenitally no sense of pain, the elderly who have no sense of pain due to cerebral dysfunction caused by dementia, and so on, to prevent sensory disturbance related to temperatures such as frostbite and low-temperature burns. There have been proposed studies that estimate the human body state, such as respiratory [1], using PPG (photoplethysmogram) sensor. However, to our knowledge, no study has been conducted to estimate the temperature of a grasped object using a PPG sensor.

2 PROPOSED METHOD

We conducted a preliminary experiment to confirm the correlation between temperature and pulse waves. Two male subjects were asked to immerse their hands in a water bath at a randomly set temperature, and their pulse waves were measured. Linear regression was conducted on the water temperature and the maxima of the pulse wave. The correlation coefficients for both subjects exceeded 0.9, confirming the existence of a correlation.

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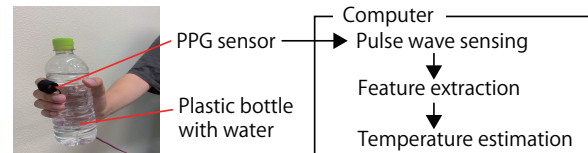


Figure 1: Overview of the proposed method.

As shown in Figure 1, the proposed method obtains pulse wave data from a PPG sensor in a wearable device. From the collected data, the maxima, minima, and amplitudes in the first 50% and last 50% windows are calculated. Five pairs were selected as feature values: 1) change rates of maxima and minima in both windows, 2) average maxima in both windows, 3) average minima in both windows, 4) average amplitudes in both windows, 5) change ratio of maxima and amplitude in both windows. A support vector machine (SVM) estimates the temperature range of an object being grasped.

3 EVALUATION

Three subjects (all males, average age 23.5 years) joined the experiment. Subjects were asked to grip a plastic bottle with water kept at a predetermined temperature for 40 seconds with a PPG sensor attached to the fingernail of the right index finger. 40-second data for four temperature ranges [0,10), [10,20), [20,30), and [40,50)°C were collected by changing the water temperature in 2°C increments, i.e., 0, 2, 4, 6, and 8°C in [0,10) range. 3 subjects × 4 ranges × 5 data = 60 data samples were collected. In this evaluation, four temperature ranges: [0,10), [10,20), [20,30), and [40,50)°C, three temperature ranges: [0,10), [10,30), and [40,50)°C, and two temperature ranges: [0,10) and [40,50)°C were tested with linear SVM and RBF kernel.

The results of feature value 5) are shown because it showed the best among the five pairs. The estimation accuracies using linear SVM were 80%, 82%, and 94% for four, three, and two temperature ranges, respectively. The estimation accuracies using RBF kernel were 72%, 71%, and 94% for four, three, and two temperature ranges, respectively. The reason the accuracy of the two temperature ranges showed high performance is that high and low temperatures have a large effect on the blood vessels, resulting in a large change in the pulse wave. In addition, the object temperature close to the body temperature caused a small change in the pulse wave.

4 CONCLUSION

In this paper, we proposed a method for estimating the temperature of an object in contact with the skin using a pulse wave. Experimental results showed 94% for the two temperature ranges.

REFERENCES

- [1] Havriushenko et al. 2020. Smartwatch based respiratory rate estimation during sleep using CNN/LSTM neural network. In *ELNANO 2020*. IEEE, 584–587.